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Paleomagnetism of Paleoproterozoic mafic dyke swarm from the Uauá region, northeastern São Francisco Craton, Brazil: tectonic implications.

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Abstract — This paper presents paleomagnetic analyses of 146 samples collected from 45 mafic dykes (sites) and 18 samples from basement rocks in contact with dykes. Two groups of dykes are recognized: the first (metamorphic dykes-MD) is composed of amphibolites with distinct schistosity and the second group is formed of two main rock types; a BD (basic dykes) type which retains its original texture and mineralogy and a MBD (metamorphic basic dykes) type where the clinopyroxene is almost completely substituted by amphibole. The MBD dykes occur near shear zones which are associated with the metamorphism of these rocks (ages of 2003 ± 86 Ma and 1975 ± 2 Ma). Most K-Ar and $^{40}Ar/^{39}Ar$ mineral age determinations on the analysed dykes and basement rocks yielded dates in the interval 1.93–2.14 Ga. Rb/Sr mineral isochrons on dykes from the BD type suggest two different generations of emplacement, at 2.38 Ga (minimum age) and 1.98 Ga. After AF and thermal demagnetization, a characteristic direction of magnetization (Dm=17.1°; Im=69.8°; N=31; K=40.6; α_{95}=4.1°) could be isolated for 29 metamorphosed and unmetamorphosed dykes, and basement rocks from two localities, which yielded a paleomagnetic pole situated at 331.4°E; 23.8°N (K=16.8; α_{95}=6.5°). The paleomagnetic, petrographic, and geochronological data are interpreted to indicate that this characteristic component of magnetization is secondary. This component may have been acquired during the uplift and cooling of the Transamazonian cycle, at about 1.98–1.9 Ga ago. Tectonically, our data when compared with paleomagnetic poles of similar age from the Kalahari Craton are consistent with a model in which the São Francisco/Congo and Kalahari Cratons were part of a single stable continental mass in the late Paleoproterozoic. © 1998 Elsevier Science Ltd. All rights reserved

Resumo — Resultados paleomagnéticos de 146 amostras pertencentes a 45 diques máficos e 18 amostras do embasamento coletadas próximo aos contatos dos diques são apresentados. Dois grupos de diques são reconhecidos na área em estudo: um grupo (diques metamórficos-MD) composto por anfibolitos com xistosidade desenvolvida e outro grupo formado por dois tipos de rochas; um tipo denominado BD (diques básicos), os quais retém suas texturas e mineralogias originais e outro tipo, denominado MBD (diques básicos metamórficos), onde os clinopiroxênios estão quase completamente substituídos por anfibólios. Os diques MBD ocorrem nas proximidades de zonas de cisalhamento, as quais estão associadas ao metamorfismo destas rochas (idades de 2003 ± 86 Ma e 1975 ± 2 Ma). A maioria das datações K-Ar e $^{40}Ar/^{39}Ar$ realizadas em minerais dos diques analisados e das rochas do embasamento forneceu idades entre 1.93 e 2.14 Ga. Isócronas Rb/Sr em diques do tipo BD indicam duas diferentes gerações, uma em 2.38 Ga (idade mínima) e outra em 1.98 Ga. Uma magnetização característica (Dm=17,1°; Im=69,8°; N=31; K=40,6; α_{95}=4,1°) pode ser isolada através dos tratamentos por campos alternados e térmico para 29 diques metamórficos e não metamórficos e para rochas do embasamento coletadas em duas localidades, a qual forneceu um pólo paleomagnético situado em 331,4°E; 23,8°N (K=16,8; α_{95}=6,5°). As evidências paleomagnéticas, geocronológicas e petrográficas sugerem que esta direção característica é de origem secundária, a qual deve ter sido adquirida durante a fase de soerguimento e resfriamento do ciclo Transamazônico, há cerca de 1.98–1.9 Ga atrás. Tectonicamente, os pólos atualmente existentes para o Craton do São Francisco, quando comparados com pólos de mesma idade pertencentes ao Craton do Kalahari, são consistentes com um modelo no qual os cratons São Francisco/Congo e Kalahari faziam parte de uma única massa continental no final do Paleoproterozóico. © 1998 Elsevier Science Ltd. All rights reserved

INTRODUCTION

The present configuration of Paleoproterozoic paleomagnetic poles from the Kalahari Craton provides a reasonable definition of that Craton's apparent polar wander (APW) path between ca. 2.05 Ga and 1.85 Ga (Morgan, 1985; Hattingh, 1989). Well-dated Paleoproterozoic paleomagnetic poles from the West African and north Amazonian cratons led Onstott and Dorbor (1987) and Renne *et al.* (1988) to suggest a movement of these two cratons relative to the Kalahari Craton during this time. For the intervening São Francisco/Congo Craton in western Gondwana (Fig. 1), however, few Paleoproterozoic paleomagnetic poles have been available. Paleomagnetic studies of Paleoproterozoic units from the São Francisco/Congo craton would be important to elucidate the paleogeographic evolution of the western Gondwana cratons at this time.

This paper presents paleomagnetic results from the Paleoproterozoic Uauá mafic dykes, situated in the northeastern part of the São Francisco Craton, Brazil (Fig. 2). A comparison with the Paleoproterozoic poles from the Kalahari Craton is presented and a tectonic model is proposed.

GEOLOGICAL SETTING

The northeastern part of the São Francisco Craton, in the Uauá region (Bahia State), contains an extensive mafic dyke swarm cutting the Uauá Metamorphic Complex (UMC, Fig. 2) of Archean age (ca. 3.1–2.7 Ga, Gava *et al.*, 1983; Jardim de Sá *et al.*, 1984; Mascarenhas

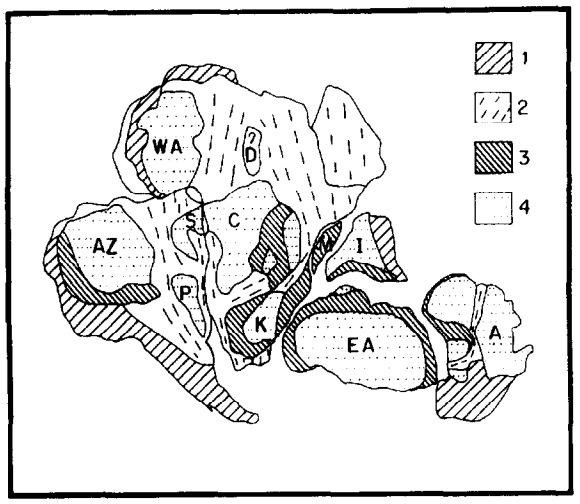


Fig. 1. Gondwana Supercontinent, probably formed by the Rodinia Supercontinent fragmentation (Hoffman, 1991; Li, & Powell, 1993). 1 — Phanerozoic belts; 2 — Pan-African belts; 3 — Grenvillian belts; 4 — pre-Grenvillian Cratons. Main cratonic units: A — Australia; AZ — Amazonia; C — Congo; S — São Francisco; D — Chad; EA — East Antarctic; I — India; K — Kalahari; M — Madagascar; P — Rio de La Plata; WA — West Africa.

and Sá, 1982; Teixeira and Figueiredo, 1991; Bastos Leal, 1992). The UMC consists of quartz-feldspathic banded gneisses, tonalitic to granodioritic ortho-gneisses, migmatites, amphibolites, ultramafic and supracrustal rocks, partially rejuvenated during the Transamazonian cycle (2.2–1.8 Ga) (Beurlen, 1970; Winge and Danni, 1980; Bastos Leal, 1992).

The area which encloses the Uauá dykes shows a sigmoidal geometric pattern outlined and limited by shear zones which affect both the rocks of the UMC and the dykes (Bastos Leal *et al.* 1994). Two groups of mafic dykes are recognized, based on field and petrographic data (Bellieni *et al.*, 1995): The first group (metamorphic dykes-MD) is composed of amphibolites with fine-grained homogeneous texture with a distinct schistosity developed before the Transamazonian cycle (Jardim de Sá *et al.*, 1984). These dykes trend N–S and have a maximum thickness of 2m. Several amphibole K-Ar ages (MD dykes) between 2.1 and 1.9 Ga are similar to ages of biotites and amphiboles in the Archean country gneisses (2.2–1.8 Ga; Bastos Leal *et al.*, 1994), which were interpreted as representing cooling ages from the final stages of the Transamazonian cycle.

Two main rock types form the second group of the Uauá mafic dykes (Bellieni *et al.*, 1995). A BD (basic dykes) type which retains its original igneous texture and mineralogy (locally a slight recrystallization may be observed in the matrix) and a MBD (metamorphic basic dykes) type, in which the original clinopyroxene is almost completely substituted by amphibole and a weak lineation sometimes exists. These dykes occur near shear zones and amphiboles from the sheared dykes (MBD type) yielded K-Ar apparent ages between 2003 ± 86 Ma and 1975 ± 2 Ma which may represent minimum ages of this event (Bastos Leal *et al.*, 1994). The BD and MDB-dykes trend mainly NE–SW (N–S and NW–SE attitudes are rare) and their thicknesses ranges from a few centimetres to 30 metres (commonly 2–3 m).

Based on Rb/Sr geochronologic age and petrographic studies Bastos Leal *et al.* (1994), proved that the BD set actually consists of two separate magmatic episodes, tentatively associated with the early- and late-stage evolution of Paleoproterozoic mobile belts developed in the northern part of the São Francisco Craton (Teixeira and Figueiredo, 1991).

Rb-Sr mineral isochrons obtained from individual dykes (BD dykes) reveal two distinct age patterns: (i) an older age with a date of 2384 ± 114 Ma (with an initial ratio (IR) of 0.7008 ± 0.0002) which was interpreted as the age of dyke intrusion; and (ii) younger ages ranging from 1983 ± 31 Ma to 2205 ± 33 Ma. Based on isotope evidence (variation of the $^{87}Sr/^{86}Sr$ initial ratios), Bastos Leal *et al.* (1994) consider the age of 1983 Ga to represent the youngest episode of dyke intrusion in the Uauá region, because the other isochrons may have been contaminated.

Finally, in the Uauá area, $^{40}Ar/^{39}Ar$ step-heating analyses on biotite from the country rock in contact with a dyke and on hornblende from an amphibolite dyke yielded age-spectra with interpreted dates about 2.0 Ga (T.C. Onstott, personal communication, quoted in Bastos Leal, 1992).

METHODS

The sampling sites (Fig. 2) included dykes from both groups of dyke intrusions. Two dykes located north from Uauá, exhibit crosscutting relationships (sites 4 and 5 of Table 1). However, both were further metamorphosed under amphibolite facies conditions.

Paleomagnetic analyses were carried out on 121 oriented hand samples and 25 oriented cores collected from 45 mafic dykes (each dyke considered as one site) in the Uauá region (Fig. 2). Fifteen hand samples and three cores from the basement rocks (UMC) near the contact (less than 6 m) of three dykes (indicated with * in Table 1) were also sampled for baked contact tests. Generally, three to four block samples or three to five cores spaced at least half a meter apart were collected from each dyke. Whenever possible, both solar and magnetic compasses were used for orienting samples. For most samples both measurements were in good agreement (<5° difference).

Three to four specimens 2.5 cm in diameter and height were cut from cores extracted from each hand sample, and two to three specimens of the same size were cut from the sampled cores. Both alternating field (AF) and thermal

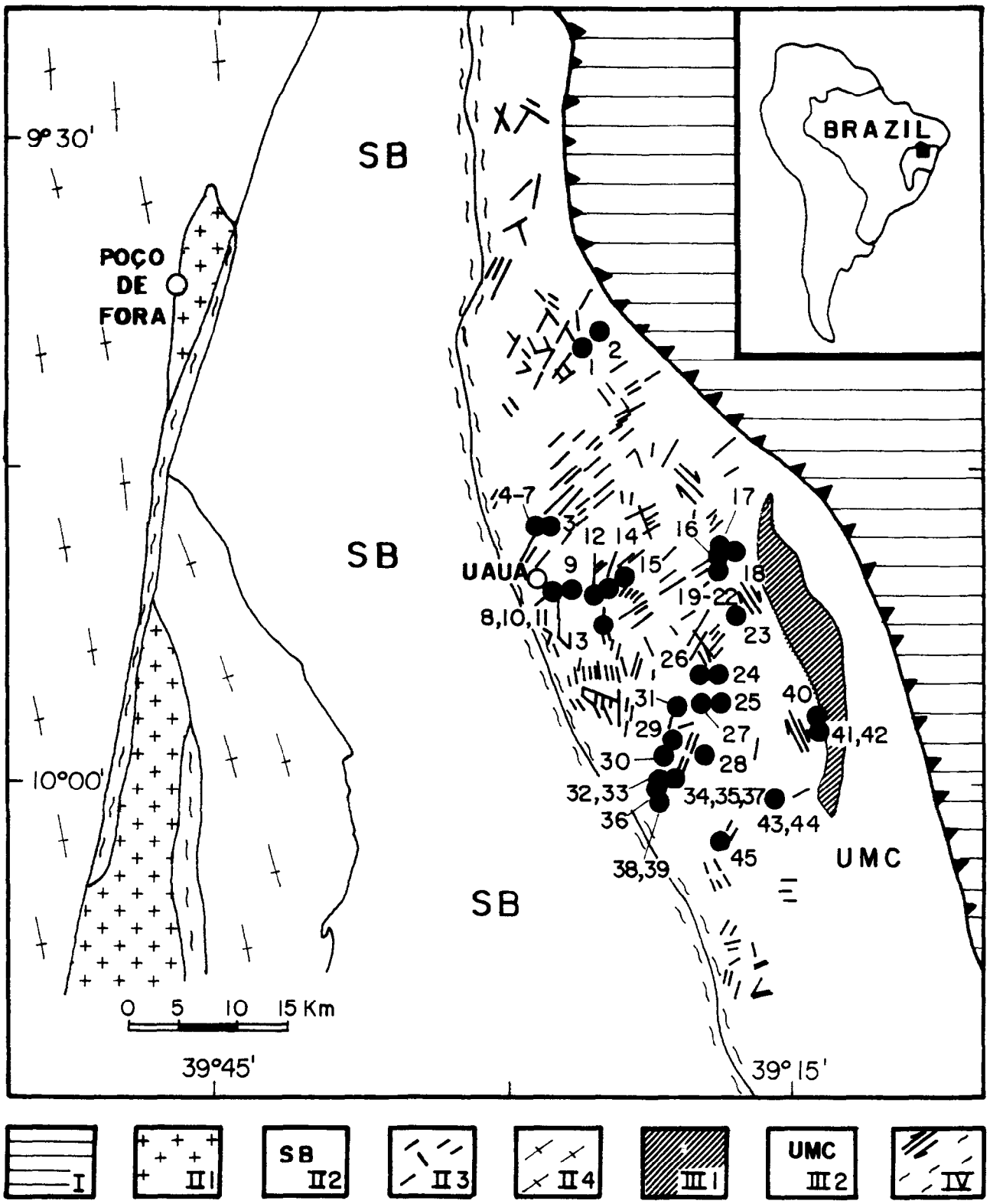


Fig. 2. Simplified geologic map of the Uauá region (Bastos Leal, 1992). Neoproterozoic: I — Sergipano fold belt, delimited by thrust faults along its western side. II — Paleoproterozoic: II1 — Itiúba sienite (1.9 Ga); II2 — Senhor do Bonfim gneissic belt; II3 — Uauá mafic dykes; II4 — Caraíba Complex. III — Archean: III1 — Supracrustal Sequence (Capim Group); III2 — Uauá Metamorphic Complex (UMC). IV — Shear zones (K-Ar ages of 2003 ± 86 Ma and 1975 ± 2 Ma). Sampling sites are indicated by full circles. (numbers from 1 to 45).

demagnetization were applied to these specimens. A spinner magnetometer was used for remanent magnetization measurements. Thermomagnetic (saturation magnetization vs. temperature) curves in air were obtained for most of the dykes by using a horizontal magnetic balance. Small amounts of crushed samples were exposed to fields between about 1.7 and 7.7 kOe, the entire heating and cooling process taking about an hour.

Magnetizations considered characteristic of the

Table 1. Paleomagnetic results from the Uauá mafic dykes

Site	Samples	P	T	TC (°C)	N	Site mean directions DEC (°)	INC (°)	α95 (°)	K	VGP PLONG (°E)	PLAT (°N)
1	BH-28 to 30	BD	c	—	2	348.5	55.4	10.1	620	307.7	43.2
2	BH-31 to 33	MBD	c	581	2	17.2	75.4	33.5	58	328.8	16.7
3	BH-11 to 14	BD	c	—	4	26.6	75.0	8.3	124	333.2	15.4
3*	BH-15 to 20	—	—	—	5	359.8	61.5	17.5	20	320.3	37.5
4	UA-8A to 8D	MD	a	591	4	3.3	69.1	7.8	138	322.8	27.5
5	UA-9A to 9D	MBD	c	—	3	25.6	69.8	7.3	288	336.7	22.9
6	BH-24 to 27	BD	c	557	2	38.6	55.2	16.2	241	357.1	31.8
7	BH-21 to 23	BD	—	—	3	348.4	65.5	11.2	121	311.3	31.6
8	BH-114 to 117	BD	a	575	4	8.3	66.6	9.4	96	326.8	30.6
9	CB-4 to 6	BD	a	575	3	24.3	69.3	8.3	223	336.2	23.9
10	BH-41 to 44	BD	a	586	4	317.4	76.4	17.2	29	303.2	9.2
11	BH-48 to 51	BD	e	579	3	25.9	73.2	6.9	318	334.3	18.2
12	BH-45 to 47	BD	e	573	3	49.5	73.6	9.8	161	343.6	10.2
13	BH-61 to 63	MBD	c	—	3	77.8	79.6	17.5	51	340.4	-5.1
14	BH-52 to 54	BD	c	579	3	359.6	67.1	13.3	87	320.3	30.3
15	BH-57 to 60	BD	e/t	584	2	36.2	73.5	13.1	366	338.7	14.9
16	BH-55	BD	c	—	1	80.2	57.6	10.6	136	11.4	1.5
16*	BH-56	—	—	—	1	27.1	-18.8	21.7	135	52.7	63.3
19	UA-1A to 1C		b	576	3	1.8	-35.6	17.7	50	130.7	80.0
20	UA-2A to 2B		a	570	1	335.8	74.1	2.9	7232	308.4	17.2
21	UA-3A to 3C	BD	a	571	3	7.7	71.7	9.5	170	325.3	23.3
22	UA-5 to 7	MBD	a	585	3	93.2	80.1	17.7	50	340.2	-10.3
24	BH-122 to 125	MBD	c	—	4	26.9	64.8	10.7	75	342.3	28.4
25	BH-118 to 121	BD	e	588	2	2.9	62.9	18.4	186	323.3	35.6
26	BH-64 to 66	BD	a	582	3	13.3	71.8	30.8	17	328.5	22.5
27	BH-77 to 80	MBD	c	—	2	30.7	65.6	3.2	6209	343.2	26.2
28	UA-10A to 10D	MBD	d	581	4	358.0	81.2	11.2	68	320.1	7.2
29	BH-81 to 83	BD	c	567	2	19.3	60.6	29.2	75	338.3	35.5
30	UA-11A to 11E	MBD	d	595	3	62.6	59.9	11.5	116	4.3	13.3
31	BH-84 to 86	BD	e	593	2	352.4	62.5	29.0	76	313.9	35.8
31*	BH-87 to 89	—	—	—	2	9.6	61.3	34.5	54	329.6	36.9
38	BH-105 to 107	MD	c	—	3	354.6	51.1	7.2	301	313.8	47.8
39	BH-108 to 110	MD	c	—	3	17.9	63.6	9.2	180	335.5	32.5

This table includes only the dykes that yielded consistent stable site directions. P (Petrography, according to Menezes, 1992): BD — basic dykes with primary mineralogy and texture, although they present some recrystallization of pyroxenes; MBD — metamorphic basic dykes, in which the original clinopyroxene is almost completely substituted by amphibole and a weak lineation exists; MD — metamorphic dykes (amphibolites) with fine-grained homogeneous texture with distinct schistosity. T (type of thermomagnetic curve — Figure 8); TC (Curie Temperature); N (Number of samples used for site mean directions); DEC (declination); INC (inclination); α_{95}, K (Fisher's (1953) statistic parameters; Obs.: although without meaning, these parameters are also shown for means using one and two samples, for comparison purposes); VGP (Virtual geomagnetic pole); PLONG (paleolongitude); PLAT (paleolatitude); *(basement rocks).

specimens/samples were identified based on demagnetization behavior as viewed in stereographic projection. The isolated site directions were those which gave the best within-sample clustering. Orthogonal demagnetization diagrams (Zijderveld, 1967) were also used to interpret the magnetic components in the rocks. Fisher's (1953) statistics were used for sample and site mean directions.

PALEOMAGNETIC RESULTS

Most of the metamorphic dykes from the first group (MD dykes) revealed a complex behavior during AF and thermal demagnetization. Many samples yielded uninterpretable demagnetization behavior. Others, although displaying interpretable demagnetization behavior, yielded inconsistent within-sample and/or within-site magnetization directions. The NRM (natural remanent magnetization) intensities of these dykes range from 10^{-4} to 7×10^{-2} A/m and the measured magnetic susceptibilities are in the 62×10^{-5} to 113×10^{-5} (SI units) range. Nevertheless, three of these dykes (sites 4, 38, and 39 in Table 1), 26 dykes from the second group of dykes (BD and MBD dykes) and some basement rocks yielded consistent magnetic behavior during AF and thermal demagnetization.

Most of the analysed samples revealed a northwest to northeast declination, steep positive inclination remanence (Figs 3 and 4, and Fig. 5). The median destructive fields (MDF) of these dykes range from 9 mT to 80 mT. The thermal treatment suggests magnetite as the main carrier of this magnetization which has, generally, high laboratory unblocking temperatures (500°C–600°C, Fig. 6).

For some samples, demagnetization revealed other magnetizations with higher coercivities and/or laboratory unblocking temperatures, as was the case for site 19 where a north directed, moderate negative inclination magnetization was isolated (Fig. 4a). To the north of Uauá where two amphibolite dykes display a crosscutting relationship (sites 4 and 5, Table 1), a north to northeast declination, steep positive inclination remanence was found in both dykes. Only two specimens from one sample from the older dyke (site 4) yielded a distinct northwest declination, shallow negative inclination remanence during demagnetization (Fig. 4b).

Samples from site 30 (cores UA-11-A to UA-11-E) displayed a different magnetic behavior. Specimens from core UA-11-A revealed a south declination, shallow positive inclination remanence after AF treatment. Thermal

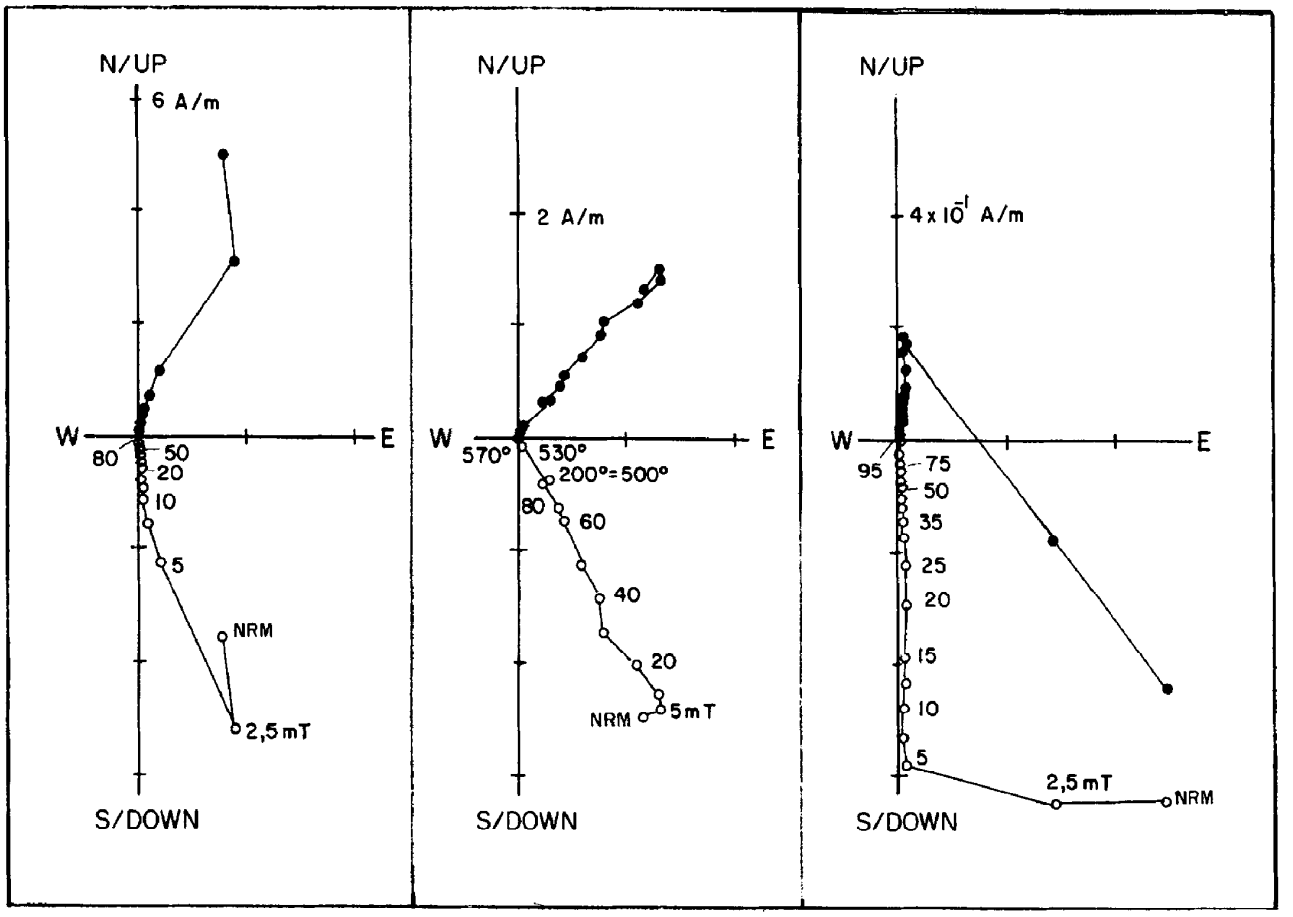


Fig. 3. Orthogonal demagnetization (AF and thermal) diagrams for specimens from three different dykes, showing the north to northeast declination, steep positive inclination, magnetization. Open (full) circles are vertical (horizontal) projections.

demagnetization for this sample isolated a distinct southeast declination, moderate to steep positive inclination magnetization not revealed in other dykes. However, specimens of cores UA-11-B, UA-11-D and UA-11-E displayed a northeast declination, steep positive inclination magnetization after treatment, similar to those obtained for the majority of the dykes. For several specimens, thermal treatment was most effective in isolating this magnetization (UA-11-E1, UA-11-D3, Figs. 5a and 5b) which has the highest laboratory unblocking temperatures, between 550°C and 600°C (see Figs. 5 and 6). Specimens from core UA-11-C (site 30) yielded uninterpretable demagnetization results.

Basement rocks (UMC) usually behaved poorly during AF and thermal demagnetization. Where demagnetization results were interpretable, basement rocks yielded magnetizations similar to those presented by the dykes themselves (Table 1). Only one basement rock (sample BH-56) sampled at site 16* (Table 1) yielded a northeast declination, shallow negative inclination magnetization, similar to that isolated for site 19 (Fig. 4a).

The characteristic site mean directions isolated for these dykes are shown in Fig. 7, and Table 1 summarizes the paleomagnetic results. There is a good clustering of directions with steep positive inclinations around the mean Dm=17.1°; Im=69.8° (N=31; K=40.6; α_{95}=4.1°). The corresponding paleomagnetic pole is situated at 331.4°E; 23.8°N (K=16.8; α_{95}=6.5°).

THERMOMAGNETIC ANALYSIS

With the exception of site 7 (weathered dyke), thermomagnetic curves were obtained for at least one sample of each dyke. Table 1 summarizes the different types of curves obtained for the dykes, represented by a letter from *a* to *f* (Fig. 8).

Some of these thermomagnetic curves (Fig. 8a) are typical of rocks with Ti-poor titanomagnetites. Others (Figs. 8b and 8f) suggest also the presence of metastable maghemite, with this mineral changing to hematite at ca. 370°C–400°C. However, several dykes presented thermomagnetic plots typical of rocks with predominance of paramagnetic phases (Figs. 8c and 8d). Nevertheless, the AF and thermal demagnetization indicate the presence of a characteristic magnetization in these rocks, suggesting magnetite as the main magnetic carrier of this component (see above). The paramagnetic minerals do not contribute to the remanent magnetization in the rocks. On the other hand, the paramagnetic phases may obscure the contribution of ferrimagnetic minerals in thermomagnetic curves, if these minerals are scarce (Housden, Sá and O'Reilly, 1988). Thin section analyses of these dykes indicate a small quantity of opaque minerals for several dykes.

To elucidate these observations, the magnetic fractions from dykes which yielded paramagnetic predominance in the thermomagnetic curves were extracted using a

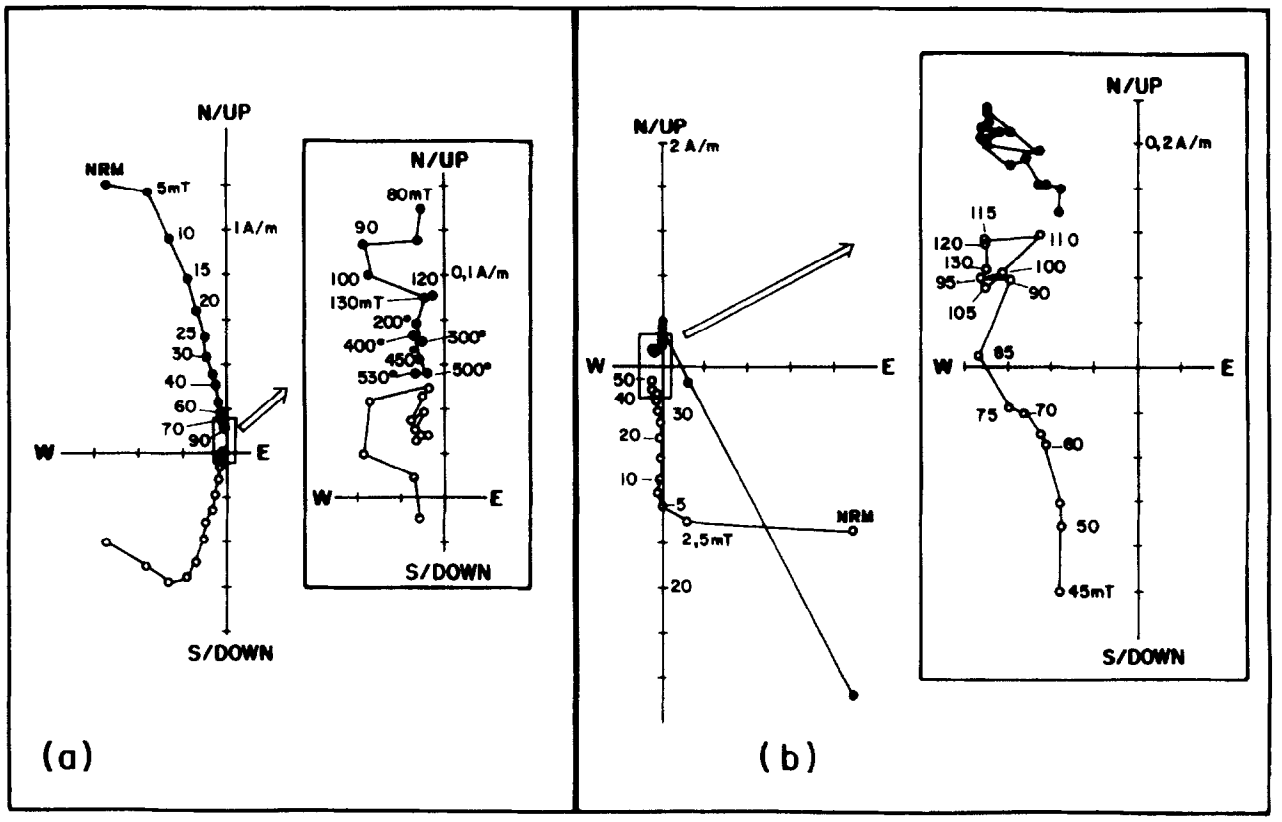


Fig. 4. Orthogonal diagrams showing the demagnetization (AF and thermal) behavior for specimen UA-1-A2 (site 19) (**a**) and specimen UA-8-A2 (site 4) (**b**). Open (full) circles are vertical (horizontal) projections.

magnet, and a thermomagnetic analysis was performed. This procedure was possible only for samples which yielded the highest intensities. Samples which first produced curves similar to those of Figs. 8c and 8d, now yielded curves typical of Ti-poor titanomagnetite minerals (Fig. 9), with Curie temperatures between 557°C and 581°C (compare results for samples UA-10-B and BH-33 in Figs. 8d, 8c, and 9a, 9c).

DISCUSSION

The characteristic direction of magnetization obtained for the analysed dykes (see above) is clearly different from the present-day dipole and geomagnetic fields (Fig. 7) and its corresponding paleomagnetic pole does not fall on any part of the late Paleozoic and Mesozoic apparent polar wander (APW) paths constructed for South America (Ernesto *et al.* 1990; Morel and Irving, 1978). The pole also differs considerably from those determined for the 1.1–1.0 Ga unmetamorphosed dyke swarms from the eastern part of the São Francisco Craton (D'Agrella-Filho *et al.*1990).

Although the Uauá pole could be interpreted as part of Neoproterozoic-early Paleozoic APW paths constructed for Gondwana (McWilliams, 1981), isotopic and geochronologic data for the Uauá dyke swarm, discussed above, suggest remanent magnetization acquisition during the late Paleoproterozoic.

The geochronologic data from the Uauá dykes and their petrographic and paleomagnetic characteristics support a secondary origin for this characteristic magnetization (at least for part of the analysed dykes), for the following reasons:

1) This magnetization was identified in dykes with different degrees of metamorphism, including two amphibolite dykes (sites 4 and 5 from Table 1) showing crosscutting relationships.
2) The opaque minerals identified in the metamorphosed dykes, generally, occur as inclusions in amphiboles of metamorphic origin.
3) AF and thermal demagnetization suggest the presence of additional distinct magnetization components with higher coercivities and/or blocking temperatures. However, as already described (see previous section), the paleomagnetic treatments isolated well defined directions for only two dykes; one dyke (site 19 of Table 1) yielded a north declination, moderate negative inclination magnetization (Fig. 4a), coherent for several specimens of this dyke and a sample from an amphibolite dyke (site 4) yielded a northwest declination, shallow negative inclination magnetization (Fig. 4b). This dyke is cut by another amphibolite dyke (site 5 of Table 1), which however, yielded only the typical north declination, steep positive inclination magnetization.
4) The presently available Paleoproterozoic paleomagnetic poles (2.1–1.8 Ga) from the São Francisco/Congo and Kalahari Cratons are presented in Table 2 and Fig. 10. South America and the São

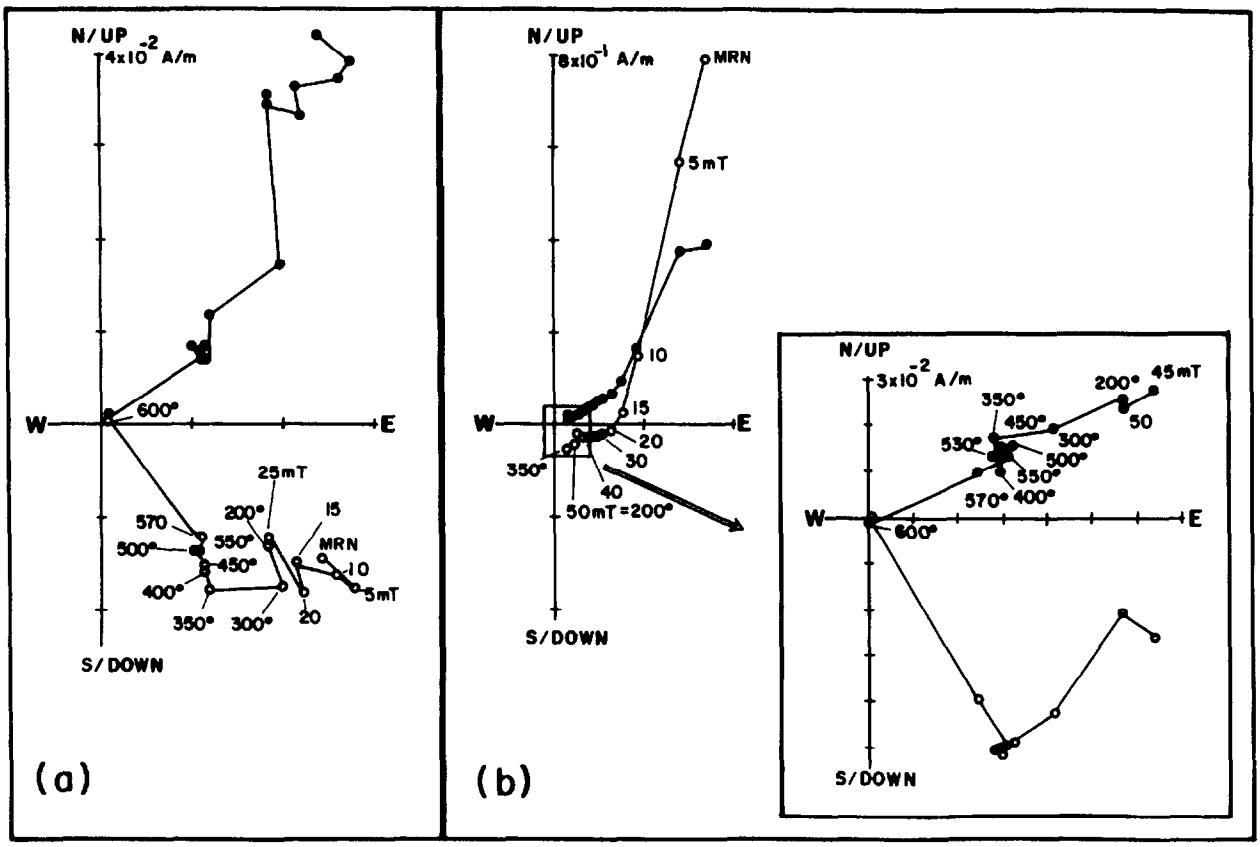


Fig. 5. Orthogonal diagrams showing the demagnetization (AF and thermal) behavior for specimen UA-11-E1 (**a**) and specimen UA-11-D3 (**b**) (site 30). Open (full) circles represent vertical (horizontal) projections.

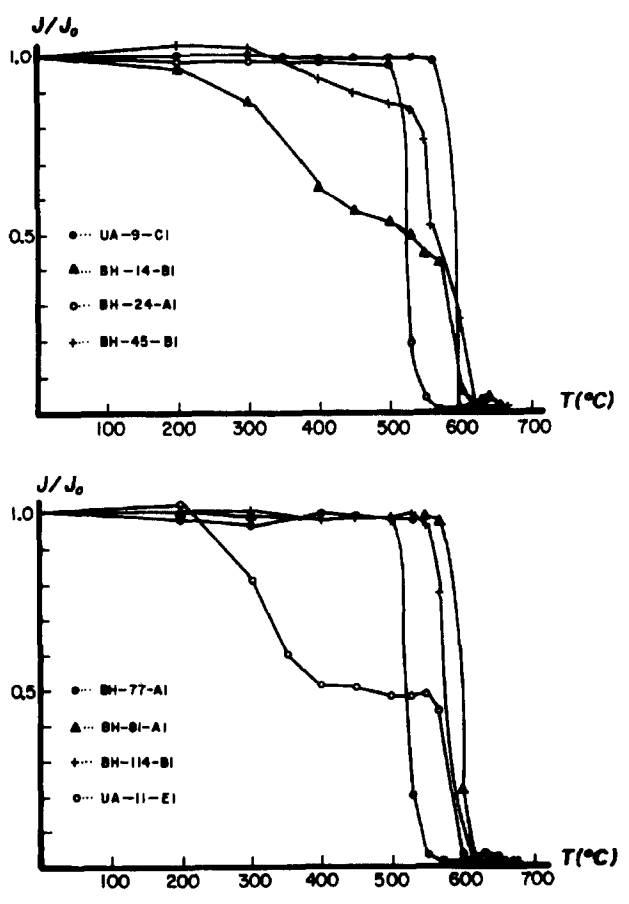


Fig. 6. Examples of normalized intensity curves during thermal treatment. Each specimen is represented by a different symbol.

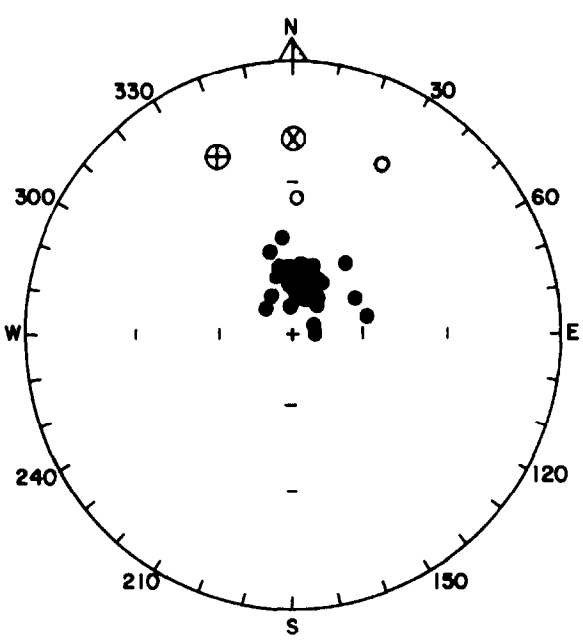


Fig. 7. Site mean directions of characteristic remanent magnetization for the Uauá dykes. Open (full) circles represent negative (positive) inclinations. The symbols ⊗ and ⊕ represent the present dipolar field and the present geomagnetic field (negative inclinations), respectively.

Francisco Craton poles (Fig. 10) were rotated to Africa in a pre-drift configuration, according to Rowley and Pindell (1989). The APW path shown in Fig. 10 is the same presented by Morgan (1985) for

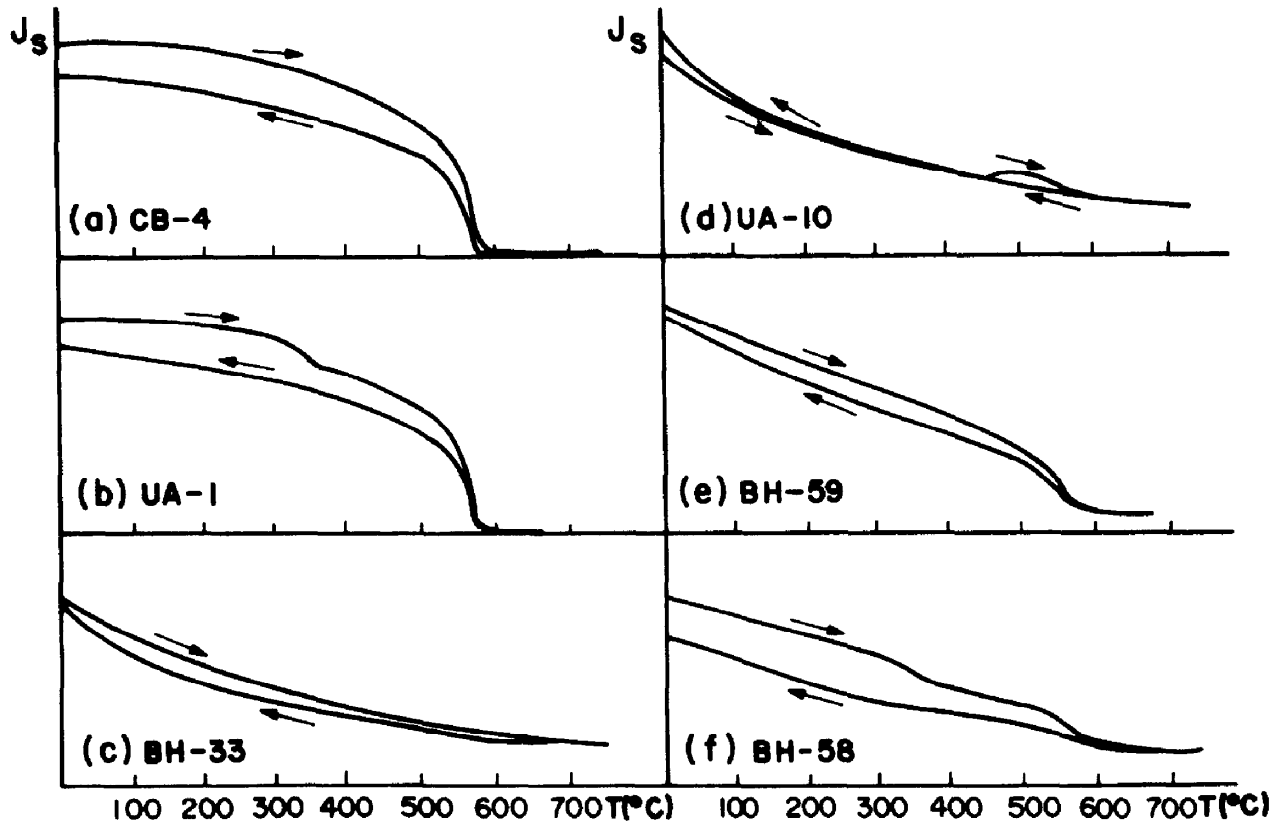


Fig. 8. Typical thermomagnetic plots for the Uauá dykes. The sample name is indicated in the figure. Table 1 presents the type of curve samples from each dyke presented (letters from a to f).

the Kalahari Craton, for the time interval between 2.05–1.8 Ga. Also plotted in this figure is the preliminary pole determined for granulites from the Jequié Complex (JC), São Francisco Craton (D'Agrella-Filho and Pacca, 1994). The granulites yielded an age of approximately 2.0 Ga by the $^{40}Ar/^{39}Ar$ method (Sletten, 1989) interpreted as the cooling of the Transamazonian cycle. This pole agrees with other age-equivalent poles from the Kalahari Craton, when South America is restored to the configuration of Gondwana. Comparison of the Uauá pole (UD, Table 2)) with poles GDE (Great Dyke extension, age of 1895 ± 74 Ma, Table 2) and LM (Limpopo metamorphics, age of 1980 ± 39 Ma, Table 2) in figure 10 suggests an age of about 1.9 Ga to 2.0 Ga for this pole, which is consistent with the age of the shearing event that is associated with the metamorphism of part of the Uauá dykes (see topic of Geological Setting). This time interval (1.9–2.0 Ga) represents also the regional cooling phase of the Transamazonian cycle.

5) Temperatures for the amphibolite facies metamorphism are in the interval 550°C and 750°C (Pullaiah *et al.* 1975). As some of the Uauá dykes exhibit amphibolite grade metamorphism, it is reasonable to suppose that the Uauá region has reached temperatures exceeding 500°C during metamorphism. The K-Ar isotopic age data obtained on different minerals (biotite and amphibole) for both the Uauá metamorphic dykes and the UMC (Bastos Leal, 1992) suggest a very low rate of cooling of the area during the final uplift stages of the Transamazonian cycle.

Based on the above considerations, we propose that the remanent magnetization found in the majority of the Uauá dykes and some basement rocks collected near dyke contacts is a secondary magnetization acquired during uplift followed by a probably slow cooling of the crust during the final stages of the Transamazonian cycle.

CONCLUSIONS

Petrographic evidence and Rb-Sr geochronological data suggest more than one generation of dyke intrusions in the Uauá area. Part of these dykes (MDB dykes) were subjected to metamorphism near shear zones associated with the final stages of the Transamazonian cycle as suggested by K-Ar isotopic age determinations on amphiboles from the metamorphosed dykes and also from the UMC host terranes (ages between 2.18 and 1.83 Ga).

Paleomagnetic analysis revealed a northwest to northeast declination, steep positive inclination characteristic magnetization for dykes of different degrees of metamorphism. Paleomagnetic, petrographic, and geochronologic/isotopic evidence support a secondary origin for this magnetization, probably acquired at approximately 1.98–1.9 Ga ago, during regional cooling of the Transamazonian cycle.

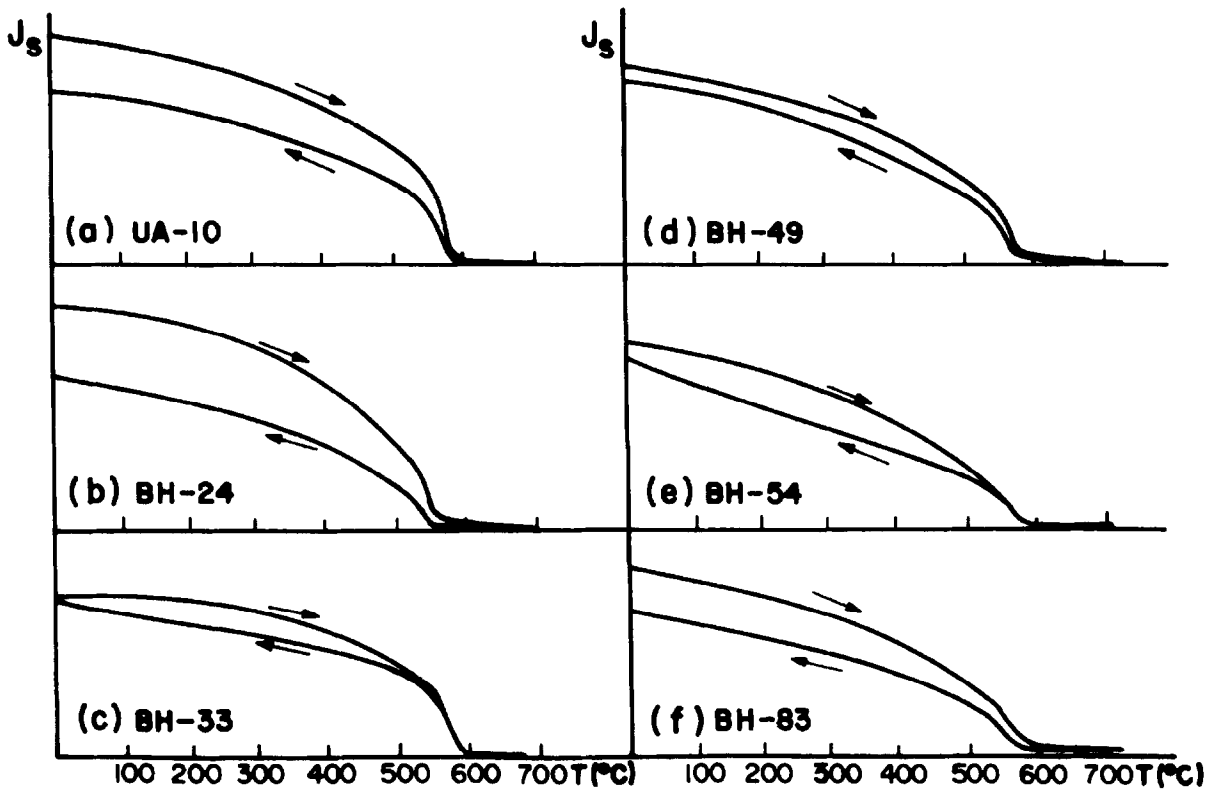


Fig. 9. Thermomagnetic curves of magnetic extracts of samples from dykes that yielded whole-rock predominance of paramagnetic phases in thermomagnetic curves. The sample name is indicated in the figure.

Table 2. Paleomagnetic poles from São Francisco, Congo, and Kalahari in the age range 2100–1800 Ma

Location rock unit	Pole	Possible age (Ma)	Plat (°N)	Plong (°E)	dp (°)	dm (A_{95}) (°)	Ref.
Kalahari Craton							
Palabora Complex	P1	2047 ± 11	35	45	7	11	1
	P2		4	356	9	11	1
Bushveld Complex							
Critical Zone	BCZ	2054 ± 11	40	47	8	13	2, 4
B Sub-Zone	BMZE		17	36	5	6	3
Main Zone	BMZW		9	27	4	5	4
Upper Zone	BUZ		16	332	15	18	5
Witwatersrand S. — overprint	Wo	1945 ± 40	19	46		(8)	6
Vredefort Complex	VC	1970–2060	27	35		(11)	7, 8
Limpopo Metamorphics	LM	1980 ± 39	26	22	8	10	9
Waterberg Sediments	W1	1720–2000	36	48	15	24	10
	W2		8	10	—	—	10
	W3		3	333	14	21	10
Rust der Winter	RDW	Waterberg	16	35	—	—	1
Sand River.s Dykes	SRD	1876 ± 68	3	9	10	11	9
Great Dyke Extension — Remag	GDE	1895 ± 74	20	0	12	15	9
Mashonaland Dolerites	MD	1850 ± 20	8	339	5	6	11,12
Congo Craton							
Kunene Anorthosite	AA	>1900	3	75	15	22	13
São Francisco Craton							
Jequié Complex	JC	1950–2050	-2	358		(18)	14
Uauá dykes	UD	1900–1980	24	331		(7)	15

(dp, dm): Semi-angles of the 95% confidence oval about the pole position; (A95): Semi-circle of 95% confidence.
References: 1 — Morgan and Briden (1981); 2 — Hattingh (1986a); 3 — Hattingh (1986b); 4 — Hattingh (1986c); 5 — Hattingh (1989); 6 — Layer *et al.* (1988); 7 — Hargraves (1970); 8 — Onstott *et al.* (1986); 9 — Morgan (1985); 10 — Jones and McElhinny (1967); 11 — McElhinny and Opdyke (1964); 12 — Compston and McElhinny (1975); 13 — Piper (1976); 14 — D'Agrella-Filho and Pacca (1994); 15 — This work.

Comparison of the UD and JC poles (Table 2) with age-equivalent poles from the Kalahari Craton (Fig. 10), suggest that the São Francisco and Kalahari Cratons were a single continental mass at the end of the Paleoproterozoic. The only Paleoproterozoic pole from the Congo Craton (Kunene anorthosite, AA in Table 2) does not coincide with the São Francisco Craton poles (Fig. 10). However, the age of this pole is yet poorly constrained (see Meert *et al.* 1994). The link between the São Francisco and Congo cratons in late

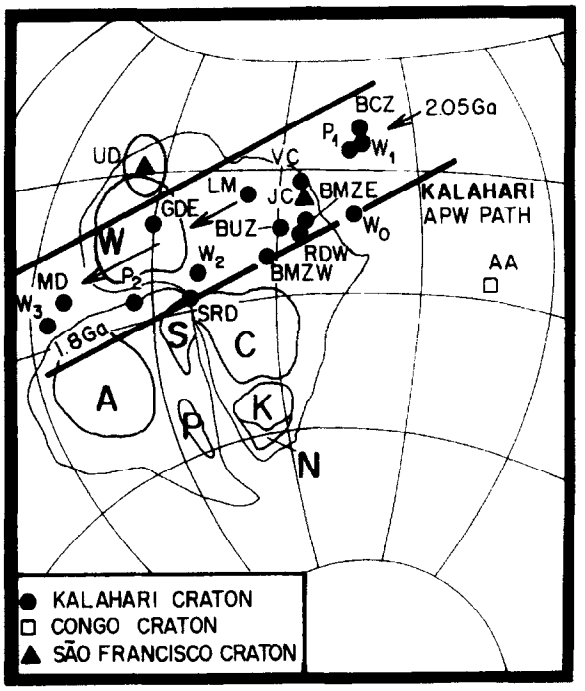


Fig. 10. Paleomagnetic poles (Table 2) from São Francisco, Congo and Kalahari Cratons for the time interval between 2.1 and 1.8 Ga. The apparent polar wander path is the same presented by Morgan (1985) for the Kalahari Craton for the time interval 2.05–1.8 Ga. The Uauá pole (UD) is ploted with its 95% confidence region. Main cratonic units: W — West African Craton, A — Amazonian Craton, S — São Francisco Craton, C — Congo Craton, P — Rio de La Plata Craton, K — Kalahari Craton, N — Namaqua Zone. Paleomagnetic poles and continents restored according to reconstruction of Rowley and Pindell (1989) (rotation pole 47.23°N, 30.84°W, 55.69° for South America to Africa).

Paleoproterozoic, however, is generally accepted based on geological evidence (e.g., Torquato and Cordani, 1981)

A possible link between São Francisco/Congo and Kalahari cratons at 2.0–1.9 Ga has an important significance for the nature of the younger Neoproterozoic Damara belt and the Rodinia Supercontinent (Hoffman, 1991). If we admit the formation of Rodinia at 1.1 Ga ago, then the São Francisco/Congo and Kalahari cratons separated sometime post 1.9 Ga and came back together in exactly the same configuration they were prior to separation. The Meso to Neoproterozoic paleomagnetic poles from the Congo and Kalahari cratons have been interpreted to favor an entirely intracratonic evolution for the Damara belt (e.g. McWilliams, 1981; Jones *et al.* 1992; Mushayandebvu *et al.* 1994), although Mesoproterozoic poles from the São Francisco Craton (D'Agrella-Filho *et al.*, 1990) have suggested a reconstruction involving separation of the São Francisco/Congo and Kalahari cratons, indicating that the intervening Damara mobile belt may record a craton-craton collision (Renne *et al.* 1990). Future new well-dated paleomagnetic poles for these two cratons are expected to elucidate this matter.

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